

# Estimating the Influence of Demographic Ageing and Migration to the Spanish Aggregate Labour Supply

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## Abstract

Population ageing has raised concerns about the future of aggregate labour supply, but its effects depend not only on the rising share of older individuals but also on immigration, participation, hours worked, and the changing age composition of the working-age population. This article estimates aggregate labour supply in Spain between 2005 and 2025 using Labour Force Survey microdata and decomposes its change by age group and birthplace status.

The results show that Spain's labour supply shifted strongly away from young workers and toward prime-age and older workers. The contribution of young workers aged 15–34 declined more than their population share, indicating weaker labour-supply attachment in addition to demographic contraction. By contrast, older workers aged 55+ increased their labour-supply share more than their population share, suggesting stronger labour-market attachment among older individuals. Immigration played a decisive compensating role: foreign prime-age and older workers generated substantial labour-supply growth, while the younger age structure of foreigners offset a large share of the negative age-structure effect associated with the ageing of the Spanish population. Overall, the findings show that ageing does not mechanically reduce aggregate labour supply; its effect is mediated by immigration, participation behaviour, and hours worked.

**Keywords:** Labour Supply, Immigration, Population Ageing, Decomposition Models.

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# 1 Introduction

Across Western Europe, Spain is a representative case of the demographic pressures currently facing developed economies. Population ageing, driven by increasing life expectancy and falling fertility rates, affects several economic and social outcomes. It places pressure on PAYG pension systems (Díaz-Giménez and Díaz-Saavedra, 2009; Boeri et al., 2024), increases public expenditure on health services (Cristea et al., 2020; Aiyar and Ebeke, 2016; Bijak et al., 2007), and raises broader concerns about social security sustainability (Faílde Garrido et al., 2021). While social security and public services often make the headlines, there is little doubt that the labour market is the first point of impact from which many of these other effects cascade.

Ageing changes the relative size of the age groups that contribute to aggregate labour supply (Maestas, Mullen, and Powell, 2023). This affects the number of potential workers, their participation rates, hours worked, and ultimately the contribution base that finances pensions, health care, and other public services. In this sense, the fiscal consequences of ageing are partly the downstream result of labour-market changes generated by demographic ageing. This article integrates the main demographic and behavioural components into a unified framework that isolates the contribution of each factor to changes in effective labour supply. The method measures aggregate labour supply as total potential annual hours worked, combining population size, age structure, labour-force participation, and hours worked by age group and birthplace status. Changes in labour supply are then decomposed into demographic, participation, and work-intensity components using a continuous-change decomposition model (Horiuchi, Wilmoth, and Pletcher, 2008).

## 1.1 Ageing and Labour-Supply Pressure

In Spain, this demographic pressure operates within a labour market already marked by structural unemployment, institutional segmentation, high sensitivity to economic cycles, the grey economy, and internal redistribution of human capital between natives and immigrants (Bentolila and Juan José Dolado, 1990; Bentolila, Cahuc, et al., 2010; Congregado et al., 2014; Güell et al., 2021; Bentolila, Juan J Dolado, and Jimeno, 2020). These features matter because they shape how demographic change is translated into labour supply. Ageing does not affect the labour market only by increasing the number of older individuals; it also changes the relative weight of age groups with different participation rates, employment prospects, and hours worked.

Direct estimates for Spain confirm this composition effect. Banco de España estimates that population ageing reduced Spain’s aggregate participation rate by about 3.4 percentage points in 2022. Without the ageing observed since 2012, the participation rate would have been 62.1% rather than 58.7%, equivalent to roughly 790,000 additional potential workers. The same study projects a further decline of about 2.8

percentage points by 2030 due to ageing, with a larger fall in the absence of net immigration (Cuadrado et al., 2023). CaixaBank Research reaches a similar conclusion: because participation follows an inverted-U pattern over the life cycle, the shift of the Spanish population toward older ages lowers the aggregate activity rate even when age-specific participation rates remain stable. It further projects that Spain’s labour force could decline by around 1.7 million people between 2016 and 2029, with the activity rate falling to 53.8%, although higher educational attainment among younger cohorts may partly offset the decline (CaixaBank Research, 2017).

More generally, international evidence shows that population ageing affects labour supply through both demographic composition and older-worker behaviour. Maestas, Mullen, and Powell show that ageing reduces labour-force growth and productivity growth, thereby lowering economic growth (Maestas, Mullen, and Powell, 2023). In the Spanish context, the OECD emphasizes that the effect of ageing depends not only on the number of older individuals but also on their ability and incentives to remain active in the labour market. Although participation among older workers has improved recently, Spain continues to face low employment rates among older individuals and institutional barriers to longer working lives (OECD, 2025). Together, these studies suggest that the labour-market effect of ageing is twofold: it reduces the relative size of high-participation age groups while making aggregate labour supply increasingly dependent on the participation, employability, and working conditions of older workers.

## 1.2 Labour force and immigrant inflows

In Western political and economic cultures, few questions are as contested, and yet as consequential, as immigration. For Spain, this debate is not abstract: immigration has become one of the forces through which the future of the labour market will be decided. After shifting from a country of emigration to a major destination in the early 2000s, Spain experienced large pre-crisis inflows that substantially rejuvenated its working-age population (Amuedo-Dorantes and De La Rica, 2008; Arango, 2013; Gonzalez and Ortega, 2011). In 2008 alone, Spain received 599,074 immigrants, with immigrants representing 13.9% of the working-age population. These inflows matter because they expand the labour force at precisely the moment when population ageing is narrowing it. In this sense, immigration appears not simply as a demographic fact, but as one of the central margins through which Spain may sustain labour supply, competitiveness, and economic growth in an ageing society (Stepanek, 2022; Okamoto, 2021; UN, 2000; Casares and Vázquez, 2018).

However, the compensating role of immigration is not automatic, since immigrant labour supply is highly exposed to economic and political cycles. The 2008 financial crisis illustrates this vulnerability: foreign workers bore a disproportionate share of job losses, immigrant employment declined, and immigration inflows from abroad

fell by more than 50% between 2008 and 2013 (Rodríguez-Planas and Nollenberger, 2016). By 2013, the immigrant share of the labour market had fallen to around 11%, and Spain experienced net migration outflows. Although immigration recovered after 2014 and foreign arrivals continued to rise after the COVID-19 crisis, these fluctuations show that immigration cannot be treated as a stable or automatic offset to population ageing. In addition, by expanding the population, migration may increase demand for public services and affect welfare spending (Fiorio et al., 2023; Naumann and Hess, 2021). Its net effect therefore depends on migrants' age composition, labour-market attachment, employment stability, and use of public services (Preston, 2014).

The key question is thus not simply whether immigration increases the size of the labour force, but whether it sufficiently increases labour supply through age composition, participation, and hours worked. On the one hand, immigration can help offset population ageing by enlarging the working-age population and increasing the number of potential contributors. On the other hand, its contribution depends on migrants' labour-market integration, exposure to economic cycles, age profile, and use of public services. The key question is thus not simply whether immigration increases population size, but whether it increases effective labour supply through age composition, participation, and hours worked.

### 1.3 The missing link: an integrated framework

Spain offers a timely and policy-relevant context for studying how demographic ageing, individual behavioural change, and immigration jointly shape aggregate labour supply. Existing studies establish that ageing and immigration are central to Spain's labour-market dynamics, but they often examine these forces separately. Research on ageing tends to emphasize pensions, health spending, social security, or the shrinking labour force (Maestas, Mullen, and Powell, 2023; Cuadrado et al., 2023; CaixaBank Research, 2017), while research on immigration often focuses on migration flows, employment outcomes, integration, or fiscal effects (Rodríguez-Planas and Nollenberger, 2016; Gonzalez and Ortega, 2011; Preston, 2014). Less attention has been given to how ageing, immigration, participation behaviour, and hours worked jointly shape aggregate labour supply.

This gap matters because demographic change alone does not determine labour supply. Ageing may reduce labour supply by increasing the relative weight of older age groups, but immigration can change the size and age composition of the working-age population, while behavioural responses can alter participation and work intensity. These channels are especially relevant in Spain, where labour-market volatility has produced different responses across groups. During the Great Recession, for example, female participation increased through the added-worker effect (Congregado et al., 2014; Sanz-de-Galdeano and Terskaya, 2020), while immigrant employment declined and reduced the foreign workforce (Rodríguez-Planas and Nollenberger, 2016). Such

patterns show that behavioural responses can either reinforce or offset demographic trends.

This article addresses that problem through an integrated framework that links age structure, participation behaviour, hours worked, and migration-driven population change. It contributes by providing a more precise measure of aggregate labour supply, based on participation rates and hours worked by age group and residence status for the population aged 15 and over, and by decomposing changes in aggregate labour supply into demographic and behavioural components. This approach is important because it identifies whether observed labour-supply changes are driven mainly by structural demographic shifts, behavioural responses, or immigration inflows, thereby moving beyond broad claims about “ageing” or “migration” and providing evidence that is more useful for policy design.

## 2 Data sources and descriptives

The dataset is built from Spain’s Labour Force Survey microdata (EPA), published by INE. Quarterly microdata files are downloaded for 2006–2025 and processed into annual estimates by age group and birthplace status for individuals aged 16 and over. Individuals are classified as Spain-born when their reported place of birth corresponds to a Spanish province, and as foreign-born otherwise. Population totals are computed using EPA survey weights. Labour-force participation is measured as the weighted share of individuals who are employed or unemployed, while working hours are measured as weighted usual weekly hours among employed workers. Annual labour input is then calculated by combining population, participation, and weekly hours, assuming a 50-week working year.

The original EPA microdata report age in grouped intervals—16–19, 20–24, and successive five-year bands up to 65+. These groups are kept when computing annual estimates and decomposition effects. However in summary tables and plots, labour supply and contributions to change, estimates are aggregated into three broader categories: 16–34 (young), 35–54 (adult), and 55+ (older). The annex contain description of the above variable related to survey questionnaire.

Table 1 reports summary indicators by birthplace status for Spain in 2005 and 2025, including population, activity rates, average weekly hours, and the share of the population aged 55 and over. The estimates are calculated from EPA microdata using survey weights, so they represent weighted population estimates rather than official population totals. They may therefore differ from official residency statistics because of differences in definitions and methodology. One important distinction is that official residency estimates are commonly reported by nationality, whereas this analysis classifies individuals by place of birth.

Table 1: Population aged 16 and over, activity, hours, and ageing indicators by birthplace, Spain, 2005 and 2025.

|                       | Birthplace status |         |          |
|-----------------------|-------------------|---------|----------|
|                       | Spanish           | Foreign | Combined |
| <b>Population</b>     |                   |         |          |
| 2005                  | 32.79             | 3.79    | 36.58    |
| 2025                  | 33.58             | 8.53    | 42.11    |
| Change                | 0.79              | 4.74    | 5.53     |
| <b>Activity rate</b>  |                   |         |          |
| 2005                  | 55.83             | 74.77   | 57.79    |
| 2025                  | 55.79             | 71.44   | 58.96    |
| Change                | -0.04             | -3.33   | 1.17     |
| <b>Weekly hours</b>   |                   |         |          |
| 2005                  | 42.63             | 42.69   | 42.64    |
| 2025                  | 39.16             | 40.11   | 39.35    |
| Change                | -3.47             | -2.58   | -3.29    |
| <b>Share aged 55+</b> |                   |         |          |
| 2005                  | 34.11             | 11.01   | 31.71    |
| 2025                  | 45.02             | 23.85   | 40.73    |
| Change                | 10.91             | 12.84   | 9.02     |

Note: Population is in millions; activity rate and the share aged 55+ are percentages; weekly hours are average hours worked per week; Combined activity rate and weekly hours are computed as population-weighted means of the age-group measures.

Table 1 shows that foreign-born individuals became a much larger part of Spain's labour supply between 2005 and 2025. Their population more than doubled, from 3.79 to 8.53 million, while the Spain-born population changed little. Foreign-born individuals also maintained higher activity rates than Spain-born individuals, although their average weekly hours declined slightly. Both groups aged over the period, with the share aged 55+ rising especially among the foreign-born population. As a result, the foreign-born share of the total population increased from 10.36% in 2005 to 20.26% in 2025.

## 3 The Labour Supply in Spain

### 3.1 Measuring the Aggregate Labor Supply

This article measures aggregate labour supply as the potential annual labour input available in the population, rather than as observed employment alone. The measure combines demographic composition, labour-force participation, and hours worked, allowing changes in labour supply to be linked separately to population ageing, immigration, participation behaviour, and work intensity. Let  $L$  denote aggregate labour supply. It is computed as the weighted sum of annual hours worked by age group  $a$  and birthplace status  $r$ , where  $r = (\text{foreign-born}, \text{Spanish-born})$ . For each birthplace group, total labour supply depends on four components: the size of the population  $pop_r$ , the age structure of that population  $str_{ar}$ , the labour-force participation rate  $wpx_{ar}$ , and annual hours worked among employed individuals  $whx_{ar}$ . Formally:

$$L = \sum_r \sum_a (pop_r \times str_{ar} \times wpx_{ar} \times whx_{ar}) = f(pop_r, str_{ar}, wpx_{ar}, whx_{ar}) \quad (1)$$

This formulation has two advantages. First, expressing labour supply in hours captures changes in work intensity that are missed by measures based only on population or labour-force counts. Second, using participation rates rather than employment alone includes both employed and unemployed individuals who are attached to the labour market. The resulting measure therefore is potential labour supply rather than realised labour demand. This distinction is important because observed employment reflects the quantity of labour absorbed by the market, but not the full amount of labour available from workers. The assumption here is that unemployed individuals are willing and available to work the average annual hours observed among employed individuals of the same age group and birthplace status. Under this assumption, equation (1) provides a consistent framework for decomposing changes in aggregate labour supply into demographic, behavioural, and work-intensity components. While this assumption make sense for demographic decompositon it's not trivial from an economique perspective, especiepecially in Spain, where long-term unemployment and skill mismatch are structural. Therefore the estimation from (1) is a potential labour supply measure, not a welfare-relevant or market-clearing concept. Table 2 summarizes the size of the population aged 16 and over and the corresponding aggregate labour supply in 2005 and 2025.

Table 2: Population and aggregate labour supply by birthplace, Spain, 2005 and 2025.

|                       | Spanish | Foreign | Combined |
|-----------------------|---------|---------|----------|
| Population in 2005    | 32.79   | 3.79    | 36.58    |
| Population in 2025    | 33.58   | 8.53    | 42.11    |
| Change, 2005–2025     | 0.79    | 4.74    | 5.53     |
| Labour supply in 2005 | 19.35   | 3.03    | 22.38    |
| Labour supply in 2025 | 18.49   | 6.07    | 24.56    |
| Change, 2005–2025     | -0.86   | 3.04    | 2.18     |

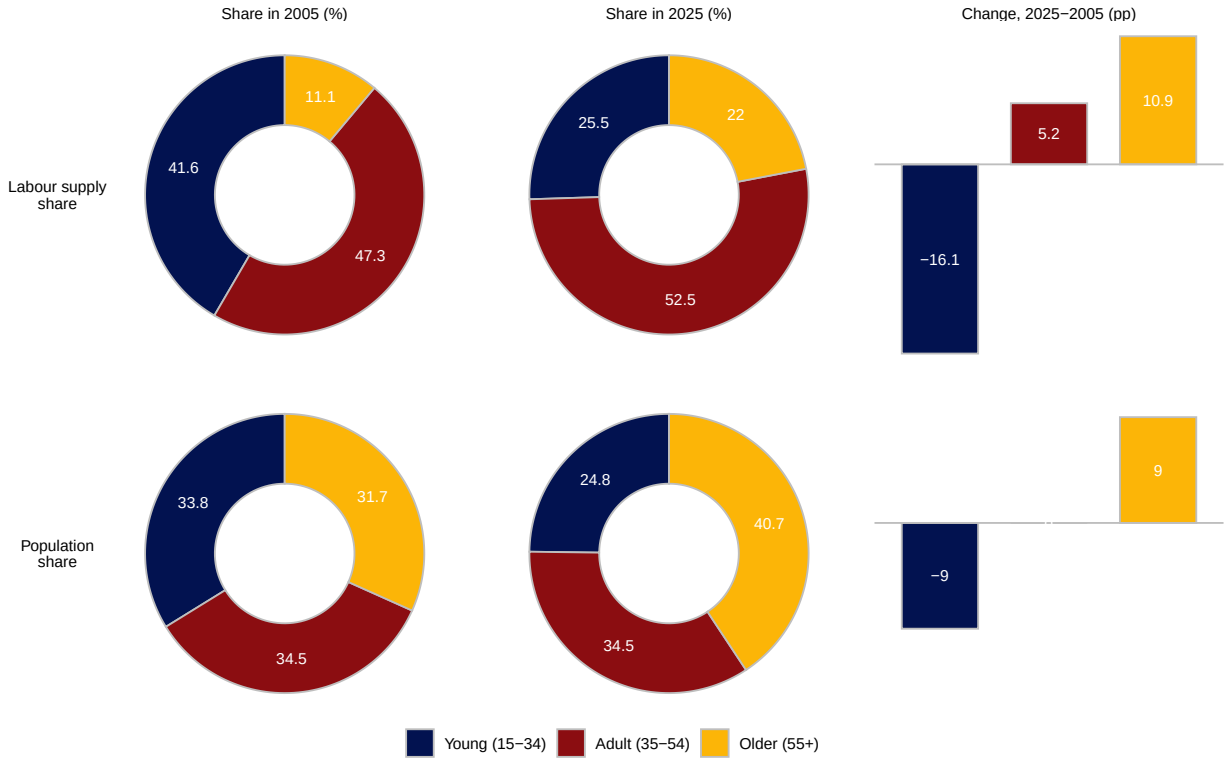
Note: Population is reported in millions. Aggregate labour supply is reported in millions of full-time equivalent (FTE) workers, where one FTE corresponds to 2,000 annual hours worked.

Table 2 provides a first overview of population and aggregate labour supply by birthplace status, comparing levels in 2005 and 2025 with the net change over the period. Between 2005 and 2025, Spain’s population aged 16 and over increased by 5.53 million, from 36.58 to 42.11 million. This growth was driven mainly by the foreign-born population, which increased by 4.74 million, compared with only 0.79 million among the Spanish-born. Aggregate labour supply rose by 2.18 million FTE workers over the same period. This increase came entirely from the foreign-born population, whose labour supply grew by 3.04 million FTE workers, while Spanish labour supply declined by 0.86 million.

This measure captures *potential* labour supply, as it is based on labour-force participation rather than employment alone. Since foreign-born individuals typically face higher unemployment rates than natives, an employment-only measure would imply a smaller and more cyclical foreign-born contribution, especially during downturns. The present approach therefore abstracts from short-run labour-demand constraints to isolate demographic and behavioural drivers of aggregate labour supply. The next subsection examines how these aggregate changes are reflected in the age composition of the population

### 3.2 Age Composition and Labour-Supply

Figure 1: Relative weight of major age groups in the population aged 16 and over and in the aggregate labor supply, and change in percentage points (pp), Spain 2005-2025

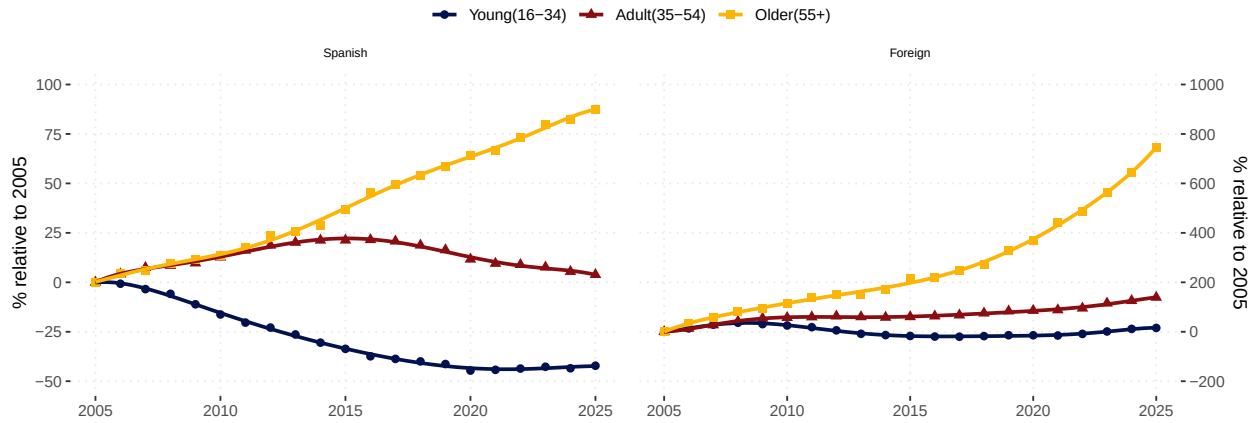


Between 2005 and 2025, Spain's labour supply shifted strongly away from young workers and toward older and prime-age workers. The young group, aged 15-34, declined from 41.6% to 25.5% of aggregate labour supply, a fall of 16.1 percentage points. This drop is much larger than their population-share decline of 9.0 percentage points, suggesting that demographic change alone does not explain the fall. Older workers, aged 55+, moved in the opposite direction. Their population share increased by 9.0 percentage points, from 31.7% to 40.7%, but their labour-supply share increased even more, from 11.1% to 22.0%, a rise of 10.9 percentage points. This implies that older individuals are contributing more to labour supply not only because they are more numerous, but also because they are more attached to the labour market than before. This behavioural change regarding labour attachment is even more prevalent among the prime-age adults whose contribution to the employment has increased by 5pp despite no change in their population share. In short, Spain's ageing population does not mechanically reduce aggregate labour supply. The figure shows that behavioural changes, especially higher labour-market contribution among older and prime-age adults, partly offset demographic ageing, while the sharp decline among young workers reflects both demographic shrinkage and weaker labour-supply attach-

ment. And to this, immigration has played a decisive role, as the next figure will show.

### 3.3 Labour-Supply Growth by Age and Birthplace

Figure 2: Trend of the aggregate labor supply as a percentage of its level at the beginning of the period, by age group for spanish and immigrants, Spain 2005-2025



The figure shows that immigration is the main source of labour-supply growth, especially among older and prime-age workers. For Spanish, young workers aged 15-34 show a persistent decline in labour supply, falling about 42.2% below their 2005 level by 2025. Prime-age Spanish aged 35-54 increased until the mid-2010s but returned close to the 2005 level by 2025, only 3.9% higher. The only strong sustained increase among Spanish is for older workers aged 55+, whose labour supply rose by 87.6%. For foreigners, the pattern is much more expansionary. Foreign prime-age labour supply increased by 138.9% relative to 2005, and foreign older labour supply rose by 745.9%. Young foreign workers are more volatile: they rose before the financial crisis, declined after 2008, and recovered to 15.5% above the 2005 level by 2025. In short, the figure suggests that Spain's aggregate labour-supply adjustment is not driven only by ageing among natives. Immigration plays a decisive role by expanding the labour contribution of foreign adults and, especially, older foreign workers, while young native labour supply continues to contract.

## 4 Decomposition of Labour-Supply Change

### 4.1 The model of continuous change: an overview

To measure the isolated effect of demographic ageing on aggregate labour supply, this article uses the continuous-change model (Horiuchi, Wilmoth, and Pletcher,

2008). This model decomposes the difference between two summary measures generated by the same process into several components, each representing the contribution of an underlying factor. The process is defined as a function  $f$ , which takes the values of the factors as arguments and returns a summary measure. A technical summary of the method is provided in the annex.

For each pair of consecutive years between 2005 and 2025, the decomposition allocates the change in aggregate labour supply using a nested three-level structure. The first level separates contributions by birthplace status, distinguishing Spanish and foreign-born populations. Within each birthplace group, the contribution is divided into four components: total population growth ( $tpop$ ), age structure ( $stx$ ), labour-market participation or activity rates ( $wpx$ ), and hours worked ( $whx$ ). Finally, within each component, the contribution is split across age groups. Each contribution piece therefore represents the change in labour hours attributable to a specific age group within a specific component and birthplace group.

A potential concern with the continuous-change model (CCM) is that it assumes smooth transitions between states, whereas Spain’s labour market was marked by sharp regime changes during the Great Recession, the migration reversal after 2008, and the COVID-19 shock. Although these episodes clearly violate economic smoothness, the CCM remains appropriate for the present analysis for two main reasons.

First, the CCM does not impose continuity in economic outcomes or behavioural adjustments; continuity applies only to the accounting path used to allocate observed changes across factors. Abrupt movements in employment, migration, or participation are therefore fully captured in the underlying data, and the decomposition integrates these discrete changes mechanically over time without smoothing or averaging away shocks. Moreover, the decomposition is implemented on a year-by-year basis and reported over sub-periods that closely correspond to major regime shifts (2005–2009, 2010–2014, 2015–2019, and 2020–2025). This effectively yields a piecewise application of the CCM, allowing crisis-period dynamics to be identified explicitly rather than diluted within longer trends.

Second, a central property of the CCM is its additivity, which holds by construction even in the presence of structural breaks. As illustrated in Table 3: the sum of the contribution pieces across all consecutive-year decompositions equals the corresponding contribution obtained from a decomposition between the first and last years. In addition, the sum of all contribution pieces equals the observed difference in aggregate labour supply between the two years, before decomposition. During crisis periods, this additivity should be interpreted strictly as an accounting identity rather than as evidence of economically smooth adjustment mechanisms. The CCM is thus used as a descriptive tool to allocate observed changes in labour supply across demographic and behavioural components, not as a model of adjustment dynamics.

Taken together, these features make the CCM a transparent and internally consistent framework for assessing how ageing, migration, participation, and hours worked contribute to labour-supply change, even in periods marked by pronounced labour-market disruptions.

## 4.2 Demographic and Behavioural Contributions

Table 3 shows the contribution of each component by birthplace and period, expressed in millions of full-time equivalent (FTE) workers, where one FTE corresponds to 2,000 annual hours worked. Positive values indicate contributions that increase aggregate labour supply; negative values indicate contributions that reduce it. The table shows that Spain’s aggregate labour supply increased by 2.18 million FTE workers between 2005 and 2025, but this net gain hides a strong contrast by birthplace. The foreign-born contribution was strongly positive, adding 3.04 million FTE workers, while the Spanish contribution was negative, reducing labour supply by 0.86 million FTE workers.

Table 3: Decomposition of changes in aggregate labour supply by birthplace, component, and period, Spain.

|                 | Period    |           |           |           |           |
|-----------------|-----------|-----------|-----------|-----------|-----------|
|                 | 2005–2025 | 2005–2009 | 2010–2014 | 2015–2019 | 2020–2025 |
| <b>Spanish</b>  |           |           |           |           |           |
| Age structure   | -2.55     | -0.26     | -0.42     | -0.76     | -1.11     |
| Activity rate   | 2.82      | 0.86      | 0.55      | 0.46      | 0.96      |
| Hours worked    | -1.60     | -0.53     | -0.31     | -0.22     | -0.54     |
| Population size | 0.46      | 0.23      | 0.15      | -0.05     | 0.14      |
| Period total    | -0.86     | 0.30      | -0.04     | -0.58     | -0.54     |
| <b>Foreign</b>  |           |           |           |           |           |
| Age structure   | -0.14     | 0.05      | -0.08     | -0.04     | -0.08     |
| Activity rate   | 0.01      | 0.08      | -0.05     | -0.06     | 0.05      |
| Hours worked    | -0.39     | -0.16     | -0.12     | -0.00     | -0.10     |
| Population size | 3.56      | 1.37      | -0.29     | 0.63      | 1.84      |
| Period total    | 3.04      | 1.34      | -0.54     | 0.53      | 1.70      |
| <b>Combined</b> |           |           |           |           |           |
| Age structure   | -2.69     | -0.21     | -0.50     | -0.80     | -1.19     |
| Activity rate   | 2.84      | 0.94      | 0.50      | 0.39      | 1.00      |
| Hours worked    | -1.98     | -0.69     | -0.43     | -0.22     | -0.64     |
| Population size | 4.02      | 1.60      | -0.14     | 0.58      | 1.99      |
| Period total    | 2.18      | 1.64      | -0.57     | -0.05     | 1.17      |

Note: Values are expressed in millions of full-time equivalent (FTE) workers.

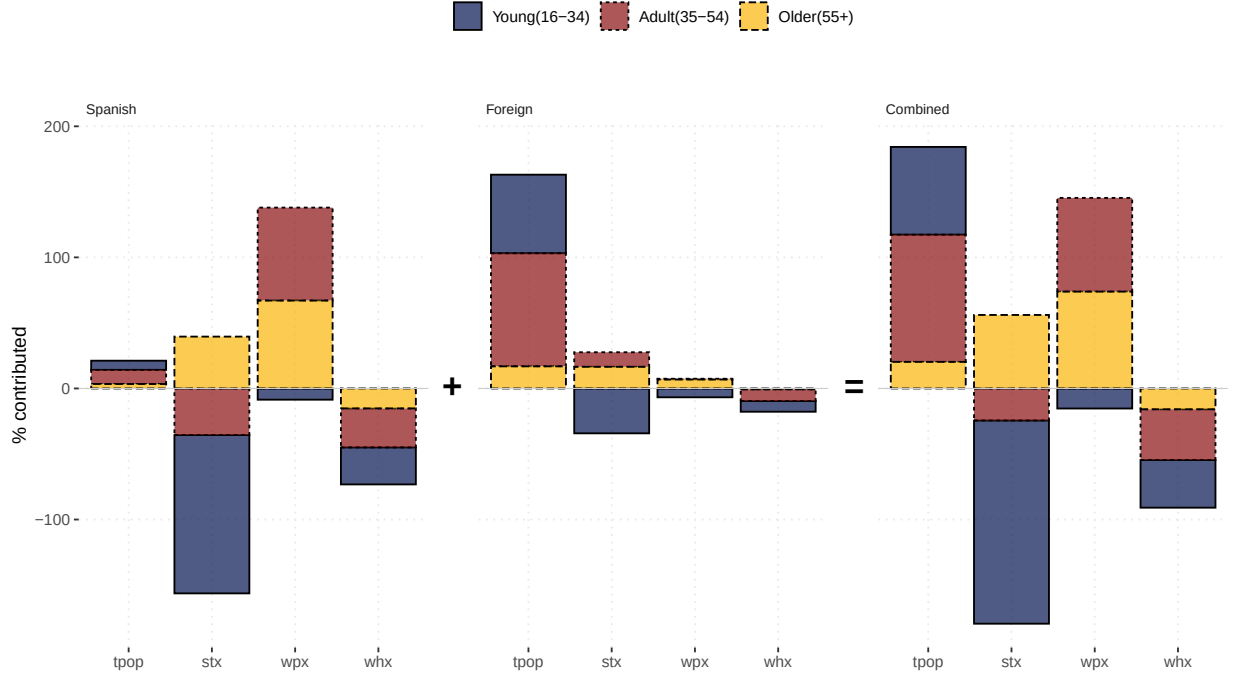
The main negative force is age structure among Spanish. For Spanish, the age-structure component reduced labour supply by 2.55 million FTE workers over 2005–2025. This confirms that population ageing among Spanish exerted a large downward pressure. However, this was more than offset by higher Spanish activity rates, which added 2.82 million FTE workers, while Spanish population size added 0.46 million. Declining hours worked reduced Spanish labour supply by another 1.60 million. For foreigners, the main positive contribution comes from population size, which added 3.56 million FTE workers. By contrast, foreign age structure slightly reduced labour supply by 0.14 million, foreign activity rates contributed almost nothing over the full period, at 0.01 million, and foreign hours worked reduced labour supply by 0.39 million. This means immigration’s contribution came mainly through foreign population growth, not through higher participation or longer hours.

Across periods, the strongest positive aggregate change occurred in 2005–2009, at 1.64 million FTE workers, and 2020–2025, at 1.17 million FTE workers. The period 2010–2014 was negative, at  $-0.57$  million, reflecting crisis-era weakness, while 2015–2019 was also slightly negative, at  $-0.05$  million. Overall, foreign population growth offset much of the negative age-structure effect among Spanish, but declining hours worked remained a drag across both groups. The relative contribution of these components across the major age groups is summarized in Figure 3.

### 4.3 Contribution of Ageing and Labour Components

Figure 3 contains three panels distinguishing Spanish, foreign-born, and combined contributions to total labour-supply change between 2005 and 2025. Within each panel, the four bars correspond to the decomposition components: total population growth (*tpop*), age structure (*stx*), activity or participation rates (*wpx*), and hours worked (*whx*). The age-group contribution pieces are grouped into three major age groups: young workers aged 16–34, adults aged 35–54, and older workers aged 55 and over. These age-group contributions are divided by the total change in aggregate labour supply, so each bar represents the percentage of total change attributable to a specific age group within a specific component and birthplace group. For the Spanish and foreign panels, the age-group contributions across all components sum jointly to 100%, because these two panels add up to the total change in aggregate labour supply. In contrast, the combined panel aggregates Spanish and foreign contributions first; within this panel, the age-group contributions across all components also sum to 100%. Positive values indicate contribution pieces that increased labour supply, while negative values indicate contribution pieces that reduced it.

Figure 3: Contribution of demographic and behavioural factors expressed as a percentage of the total variation in aggregate labour supply, Spain 2005–2025



By far, the largest negative contributions come from age-structure changes among younger and prime-age groups. The strongest drag is the age-structure effect for young Spanish aged 16–34, which contributes  $-120.8\%$  of the total change, followed by the age-structure effect for Spanish adults aged 35–54, at  $-35.6\%$ . The age-structure effect for young foreigners is also negative, at  $-34.2\%$ . In contrast, the largest positive contributions come from foreign population growth among adults aged 35–54, which contributes  $86.3\%$ , and from higher activity rates among Spanish adults and older workers, contributing  $70.9\%$  and  $67.1\%$ , respectively. Thus, the main offset to demographic ageing within the Spanish-born population comes from foreign prime-age population growth and stronger labour-market attachment among Spanish adults and older workers.

## 5 Implication discussion

The results show that immigration has played a decisive role in offsetting the labour-supply losses associated with population ageing, but they also point to the limits of relying on immigration alone. Ageing affects labour supply not only through population structure, but also through behavioural responses, health, hours worked, and exposure to labour-market shocks. The following subsections therefore discuss three

implications of the results: the extent to which immigration compensates for ageing, the potential role of healthy ageing as a complementary strategy, and the risks created by future economic and technological disruptions.

## 5.1 Compensating for ageing and the volatility trap

The age-structure effect among Spanish reduced aggregate labour supply by 2.55 million FTE workers over 2005–2025. Immigration more than compensated for this loss: the total foreign-born contribution added 3.04 million FTE workers, equal to about 119.4% of the negative Spanish age-structure effect. In short, without immigration, Spanish ageing would have reduced labour supply substantially; with immigration, the aggregate effect becomes positive. However, this migratory buffer exhibits significant pro-cyclical volatility. As evidenced in the 2010–2014 sub-period (Table 3), the foreign-born contribution turned sharply negative (−0.54 million FTE). This reveals a “volatility trap”: the primary mechanism for offsetting long-term ageing tends to stall or reverse precisely when the economy faces structural stress (Gálvez-Iniesta, 2022; FUNCAS, 2024).

The role of immigration should also be assessed in light of alternative adjustment mechanisms, especially the behavioural changes induced by ageing. Contrary to migratory buffers, behavioural adjustments among the Spanish-born population proved more resilient to economic cycles. Higher activity rates added 2.82 million FTE workers, which, despite a reduction of 1.60 million due to declining hours, resulted in a net positive behavioural contribution of 1.23 million FTE. Thus, behavioural adjustment helped offset about 48.1% of the negative age-structure effect among Spanish. While immigration remains the dominant quantitative offset, its reliability as a stabilizer is contingent on economic growth. This highlights that the requirement for migration to balance ageing could be lowered if policy shifted toward supporting the more stable margin of healthy ageing, potentially decoupling labour supply from the boom-bust cycles of migratory flows (Strzelecki, Growiec, and Tyrowicz, 2025).

## 5.2 Investment in healthy ageing as a structural stabilizer

The importance of healthy ageing as a stabilizer is underscored by recent cross-country evidence. Using data for populations aged 50+ in 41 countries from 2000–2022, Gruss et al. (2025) find that a decade of cognitive health gains raises labour-force participation by about 20 percentage points, weekly hours worked by around 6 hours, labour productivity by roughly 30%, and annual labour earnings by about 35%. In the Spanish context, such investments are increasingly critical not just for the native population, but for a foreign-born population that is ageing more rapidly than in previous decades.

The transition from the “exceptional” migratory wave of the early 2000s (Arango,

2013) to a more settled demographic has altered the foreign-born contribution profile in the 2020s (Cuadrado et al., 2023; CaixaBank Research, 2017). Between 2005 and 2025, the share of foreigners aged 55+ increased from 11.01% to 23.85%, a rise of 12.84 percentage points. This demographic maturation has led to a decline in labour-market intensity: the foreign activity rate fell by 3.33 percentage points (from 74.77% to 71.44%), and average weekly hours dropped by 2.58 hours. Without targeted investments in healthy ageing for this settled population, the “migration buffer” may become increasingly inefficient, as the volume of new arrivals would need to rise exponentially just to compensate for the declining intensity of the existing, ageing foreign-born stock (Cuadrado et al., 2023; CaixaBank Research, 2017).

### 5.3 Disruption on the horizon

Immigrants are not only ageing more rapidly than the Spanish-born population, but they are also more exposed to labour-market volatility. Foreign workers were disproportionately affected by the 2008 financial crisis: immigrant employment declined, immigration inflows fell by more than 50% between 2008 and 2013, and Spain experienced net migration outflows (Rodríguez-Planas and Nollenberger, 2016). During COVID-19, the unemployment gap between foreign-born and native workers increased from 6.7 to 10 percentage points by late 2020 (CentAUR, 2021; Fasani and Mazza, 2023; Fundación “la Caixa”, 2021). These facts do not contradict the positive demographic contribution of immigration; rather, they show that immigration is more vulnerable to labour-market disruption. This vulnerability also raises a broader question about how ongoing AI disruption may affect labour-market outcomes for immigrants and for the population more generally. At this stage, the evidence remains fragmented and inconclusive (Pizzinelli et al., 2023). Some studies find only small effects, or no clear effect, on labour demand in AI-exposed occupations (Hartley et al., 2024; D. Ghosh, R. Ghosh, Roy Chowdhury, et al., 2025). However, because AI has been shown to raise productivity in several work settings (Dell’Acqua et al., 2024; Noy and Zhang, 2023), future increases in output may require fewer additional workers than in the past. If so, Spain may not need to rely on immigration to expand labour supply to the same extent as it did over the last two decades.

## 6 Conclusion

Overall, population ageing substantially reduced Spain’s potential labour supply, but behavioural adjustment offset much of this loss. Higher participation, especially among Spanish adults and older workers, helped cushion the demographic shock, although declining hours worked limited the size of this compensation. Immigration filled the remaining gap by expanding labour supply mainly through population growth and age composition. However, this contribution depends on migrants’ labour-

market integration, sectoral exposure, and employment stability (Preston, 2014; Fiorio et al., 2023; Naumann and Hess, 2021), making it vulnerable to economic and technological disruptions. For the coming decades, complementary strategies may therefore be needed. In particular, investment in healthy ageing appears promising because it can support longer working lives and strengthen labour-market attachment among older workers in both the Spanish-born and foreign-born populations.

## 7 Annexes

### 7.1 Decomposition by Continuous Change

Horiuchi, Wilmoth, and Pletcher (2008) demonstrates that as the factors transition from values  $x_i(t_1)$  to  $x_i(t_2)$  between two times  $t_1$  and  $t_2$ , the summary measure changes from  $f(x_i(t_1))$  to  $f(x_i(t_2))$ , and the difference between these two measures can be decomposed into additive components  $c_i$  representing the contribution of changes within each factor to this difference. This is expressed by equation (2).

$$f(x_i(t_2)) - f(x_i(t_1)) = \sum_i c_i \quad (2)$$

With  $i = 1, 2, \dots, n$  being the factors involved. The decomposition is based on the assumption that the change in the covariate  $x_i$  occurs continuously or gradually over a real or hypothetical time dimension  $t_1 \rightarrow t_2$ . This provides a reasonable justification for the additivity of the effects of the factors and the elimination of interaction terms even if the process in question is a non-additive function of the covariates (Horiuchi, Wilmoth, and Pletcher, 2008, p. 786).

In other words, the continuous change model (CCM) introduces a decomposition function  $g$  which, using a process function  $f$ , transforms two series of covariates, the vectors  $X(t_1)$  and  $X(t_2)$  representing the values of the covariates  $X = (x_1, \dots, x_i, \dots, x_n)$  at times  $t_1$  and  $t_2$ , into a series of contributions or effects, the vector  $C = (c_1, \dots, c_i, \dots, c_n)$ . Thus, we can write:

$$C = g(X, f) \quad (3)$$

The series  $X$  and  $C$  having the same lengths, results in a perfect correspondence between the factors and the effects, which can be grouped for analysis purposes. Furthermore, the idea of continuity implies that each element  $c_i$  of the series  $C$  is obtained by integration according to the equation:

$$c_i = \int_{x_i(t_1)}^{x_i(t_2)} \frac{\partial f(t)}{\partial x_i(t)} dx_i(t) \quad (4)$$

In practice, equation (4) is estimated by numerical integration where  $\int_{x_i(t_1)}^{x_i(t_2)} d(t)$  is approximated by  $\sum_{x_i(t_1)}^{x_i(t_2)} \delta(t)$ , with  $x_{it} = x_i(t)$  being the value of a factor  $i$  at a moment  $t$ . Adapting the continuous change model to measure equation (1) allowed the total variation in the aggregate labor supply between 1981 and 2016 to be distributed not only among these three components but also by age group and residence status (immigrants and natives). For this article, the decomposition was performed by creating an auxiliary function  $g$  around the R package (R Core Team, 2018) DemoDecomp (Riffe, 2018). The code and data used are available upon request.

The CCM model is a better alternative to the standardization method often used in demography to account for differences in population structure when comparing demographic measures. Indeed, standardization requires a third population to serve as a standard, making it somewhat less robust as different standards can lead to different results. The CCM model solves this problem and allows for direct decomposition without interaction terms between the different factors.

Interaction effects in regression analysis refer to situations where the impact of one variable on a dependent variable is not constant or additive but depends on the values of other variables. In traditional decomposition methods, such as standardization, any discrepancy between the overall change and the sum of the individual effects of the variables is also called an interaction effect. In such cases, the interaction represents an incomplete separation of the contributions of individual covariates to the overall change or difference in a dependent variable.

## Data Source and Variables

The analysis uses microdata from the *Encuesta de Población Activa* (EPA), Spain's quarterly Labour Force Survey produced by the National Statistics Institute (INE). The EPA is a nationally representative household survey covering the population aged 16 and over. It follows a stratified, two-stage sampling design, with census sections as primary sampling units and dwellings as secondary units. Each quarterly wave includes approximately 46,000 dwellings and more than 100,000 individuals. INE provides calibrated survey weights to ensure consistency with population totals by sex, age, and region. The microdata contain detailed demographic characteristics, labour-force status, employment conditions, hours worked, and place of birth. The original variables used from the microdata files are:

- **FACTOREL (survey weight)**: Expansion factor assigned to each respondent, derived from the sampling design and calibrated to external population benchmarks. It is used to compute weighted totals and representative aggregates.
- **EDAD1 / EDAD5 (age group)**: Age information is provided in grouped intervals. Across releases, the relevant age-group variable appears as either *EDAD1*

or *EDAD5*, but both use five-year age groups for the analytical period. The analysis uses whichever variable is available and standardises it into a single age-group variable.

- **AOI (labour-force status):** Constructed by INE from responses on work activity, job search, and availability. *AOI* classifies individuals as employed, unemployed, or inactive. It is used to identify employed persons and the active population.
- **HORASH (effective weekly hours worked):** For employed respondents, the EPA records actual hours worked in the reference week, including overtime and excluding absences. These responses are coded into *HORASH* and used to compute average hours and total hours supplied.
- **PRONA1 (province or country of birth):** Respondents report their place of birth, which INE codes as Spanish provinces or foreign countries. This variable is used to distinguish Spanish-born from foreign-born individuals.

## 7.2 Trend in Labour force components

Figure 4: Average participation rate by major age groups for immigrants and natives, Spain 2005-2025

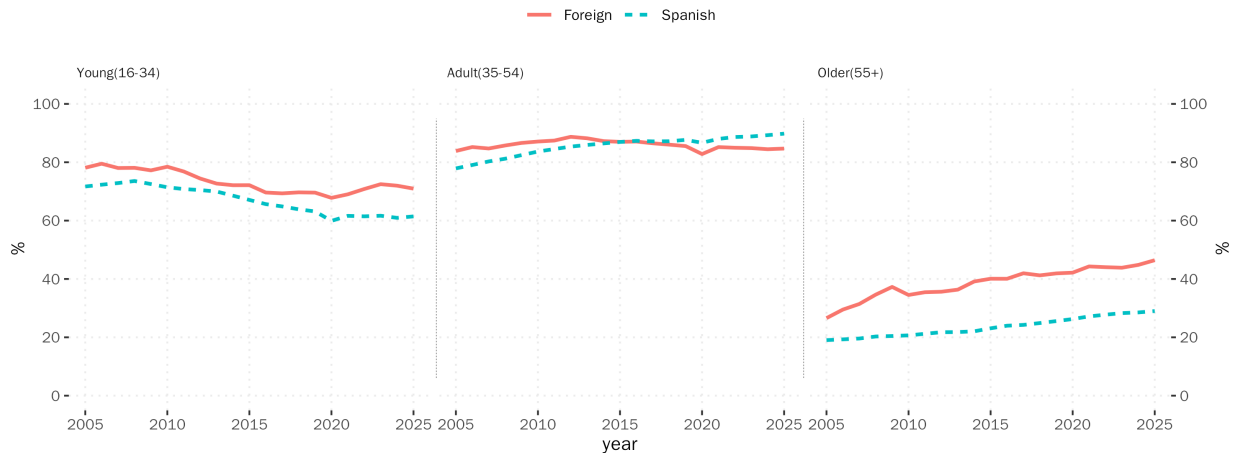


Figure 5: Average worked hours by major age groups for immigrants and natives, Spain 2005-2025

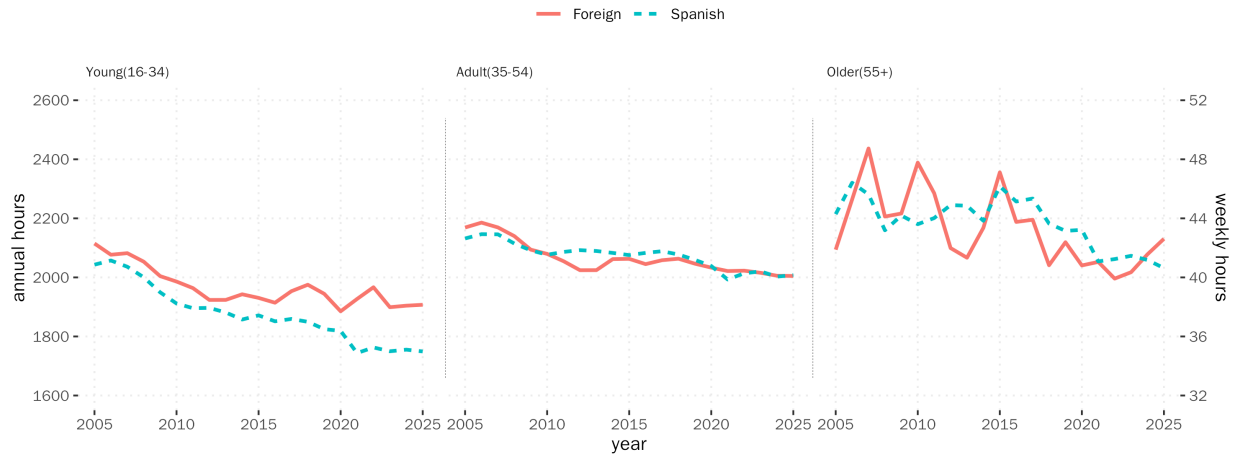
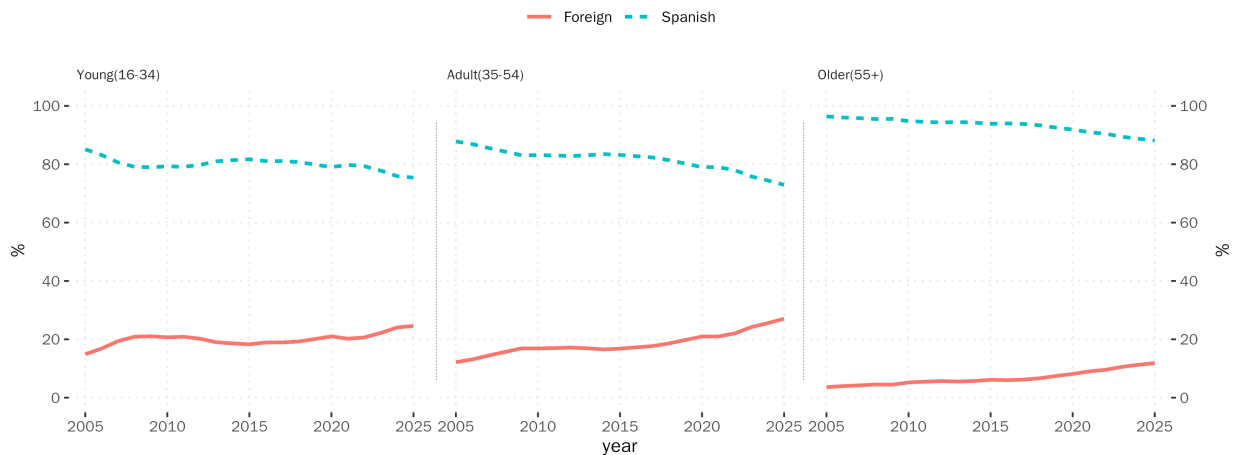


Figure 6: Percentage of total population by major age groups for immigrants and natives, Spain 2005-2025



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